

Physical Awareness for Generated Humanoid Motion

A test-time physical audit and repair loop for humanoid reference motions

CS348K Final Project

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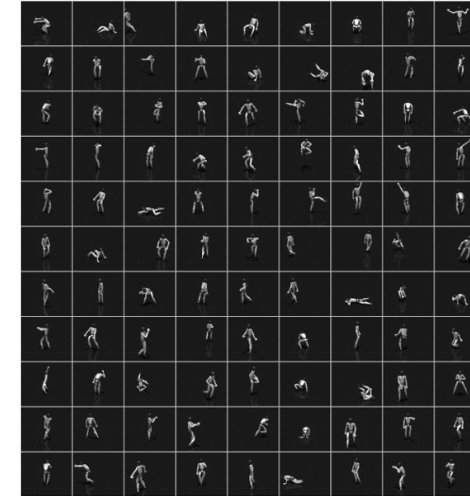
Major Advances in Humanoid Motion Tracking

Modern humanoid trackers can follow high-quality motion clips in physics.

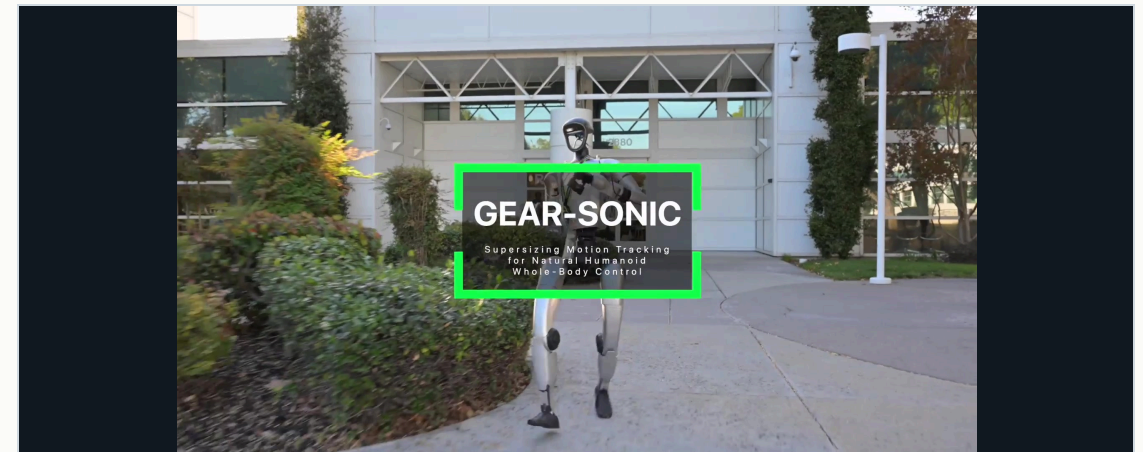
SONIC trains on over 100M frames / 700 hours of high-quality human motion.

The paper describes these motions as retargeted to the humanoid before tracking.

Tracking is getting strong. Where do these reference motions come from?



SONIC motion dataset samples diverse human motions retargeted for humanoid tracking
Source: Luo et al, Fig. 7.



KIMODO Makes References Easy To Generate

KIMODO generates G1 reference motions from text and kinematic constraints.

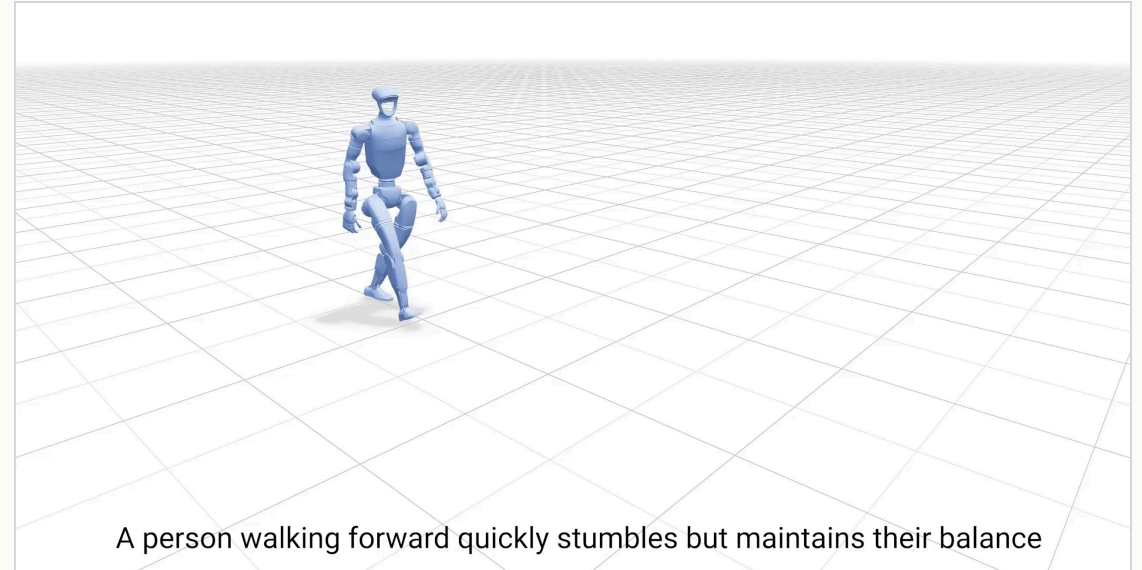
It is trained on 700 hours of production-quality optical mocap.

This gives us a broad 100-prompt humanoid reference-motion suite.

But kinematic generation does not by itself guarantee:

- feasible torques,
- valid support contacts,
- prompt correctness,
- trackability by a learned controller.

We insert a physical-awareness screen and repair loop after KIMODO and before tracking.



What Exactly Are We Refining?

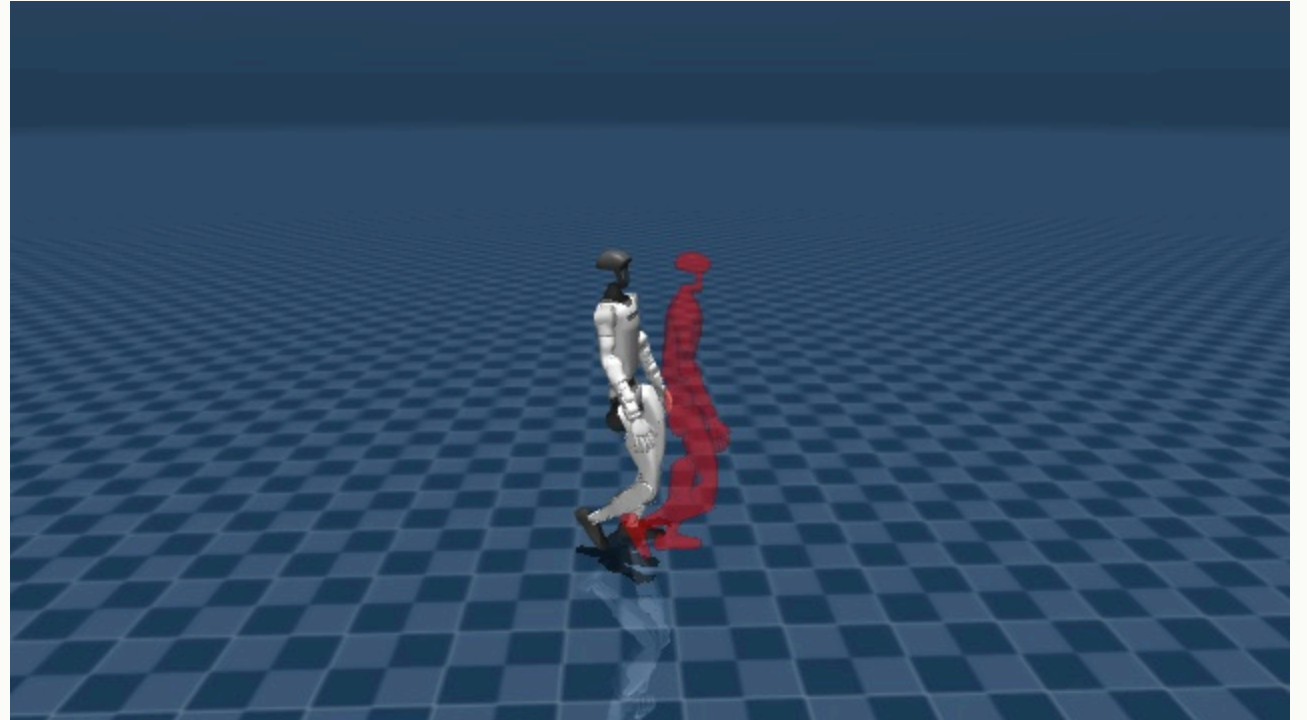
We refine generated references using failures observed in the G1 trajectory.

Each clip contains root pose and joint angles over time.

From each generated clip, we derive:

- velocity and acceleration,
- contact diagnostics,
- SONIC tracking diagnostics.

A prompt can produce a plausible replay and still need a safer reference variant.



Local G1 example: the reference replay looks plausible, but the physics rollout becomes unstable. This failure tag tells the repair loop what kind of variant to try next.

Evaluation Questions

Success has to be measured twice: first on the reference, then in the controller.

We ask four questions:

- How many first-pass KIMODO references pass physical rules?
- Which failure tags should drive repair attempts?
- Do refined or lower-risk references survive longer under SONIC?
- Which clips pass both physical rules and controller rollout?

Metric definitions:

- **physical pass:** torque/root-wrench, contact, and support thresholds,
- **SONIC no-fall:** full-horizon controller rollout,
- **RMSE:** joint tracking error,
- **failure tag:** structured context for reference repair.

What We Built

Baseline

- one generated clip per prompt,
- KIMODO G1 reference trajectory,
- no test-time refinement loop.

Our Layer: Screen + Repair

- generate a KIMODO reference,
- check dynamics/contact/support rules,
- run SONIC for rollout evidence,
- try retiming/smoothing repair variants,
- accept, flag, or reject the reference.

Method: Physical-Awareness Loop

1. Generate

KIMODO reference from the current text prompt

2. Evaluate

rules on the reference plus SONIC rollout checks

3. Repair

retime and smooth candidate reference variants

4. Gate

accept, flag, or reject after rescore

No generator or controller fine-tuning here: physics rules and SONIC decide whether a reference variant passes.

Repair: Metric-Guided Variants

When a candidate fails, the failed metric tells us what kind of reference variant to try.

Then we rescore the variants with the same physical rules and SONIC gate.

Failure tag	Repair pressure before rescoreing
high torque / root wrench	slower, smoother limb motion; avoid abrupt arm swings or snaps
self-contact	keep arms away from torso; avoid crossing legs or shoulders
non-foot floor contact	use feet-only support unless the task explicitly says crawl or kneel
support proxy	wider stance; planted support foot; smaller center-of-mass shift
contact artifact	clean foot placement; avoid dragging or sliding contacts
trackability failure	reduce speed and amplitude; make transitions gradual

■ The repair stays interpretable: every new candidate is tied to the metric that failed.

100-Prompt Humanoid Suite

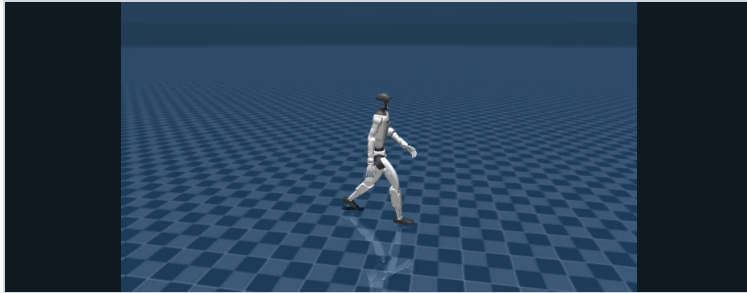
KIMODO can attempt the full prompt suite directly on the G1 skeleton.

We use concrete text prompts, not just motion labels.

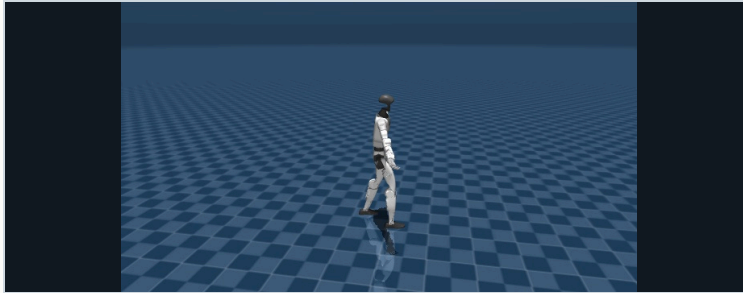
Benchmark caveat: some prompts are beyond current tracker capability, so failure is useful evidence rather than just a bad generation.

Family	Count	Example text prompts
locomotion + recovery	26	"Walk forward at a comfortable indoor pace."
manipulation + loco-manipulation	26	"Step forward, reach with the right hand as if opening a door."
floor / low posture	12	"Crawl forward using hands and feet."
dance / expressive	12	"Do an energetic happy dance with bouncing knees."
athletic + terrain stress	16	"Perform a small split-squat jump."
communication / safety	8	"Stand still and wave the right hand at shoulder height."

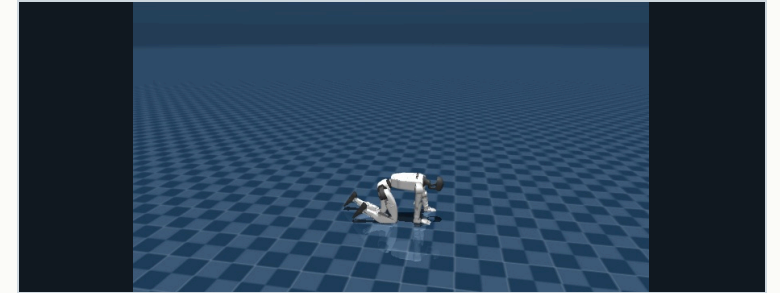
Generated Reference Examples



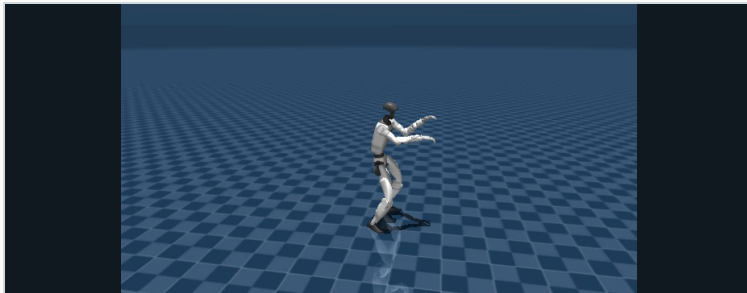
locomotion + recovery
forward_walk



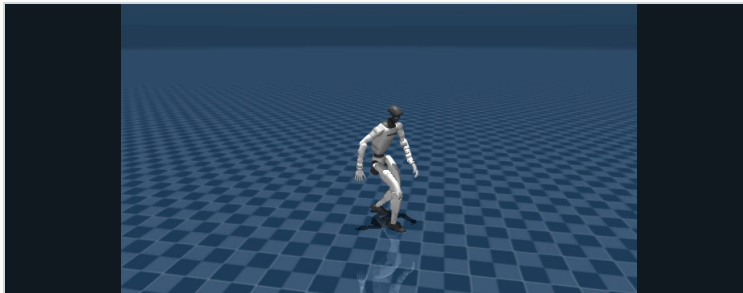
manipulation + loco-manipulation
open_door



floor / low posture
hand_crawl



dance / expressive
happy_dance



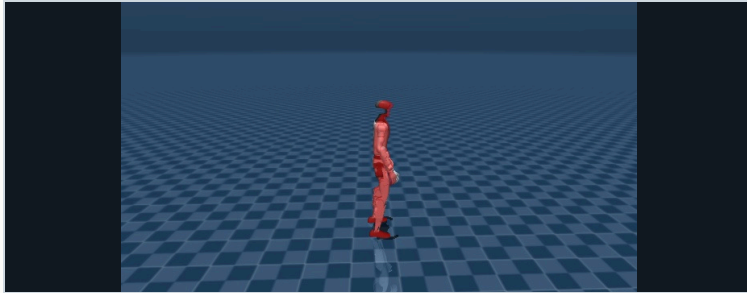
athletic + terrain stress
split_squat_jump



communication / safety
wave

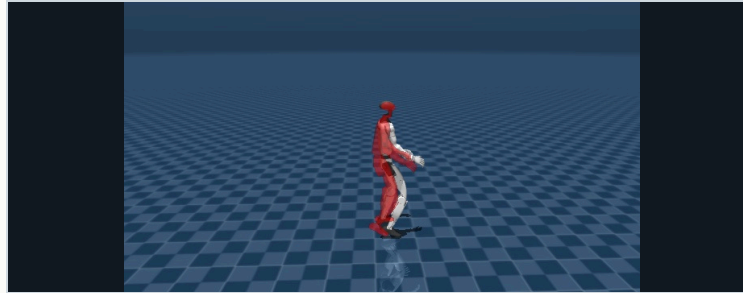
Reference-only KIMODO outputs rendered on the G1 mesh. No physics rollout is shown on this slide.

First-Pass References Can Already Track



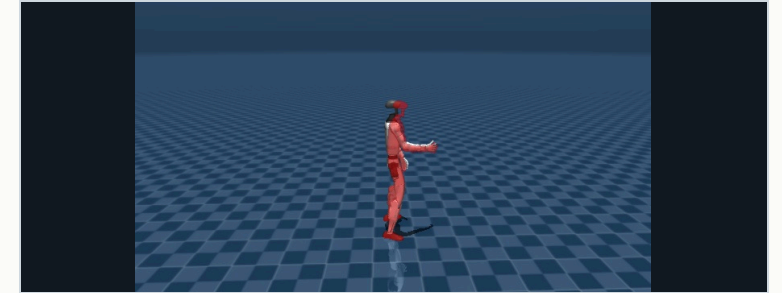
broad_jump

physical pass + SONIC no-fall



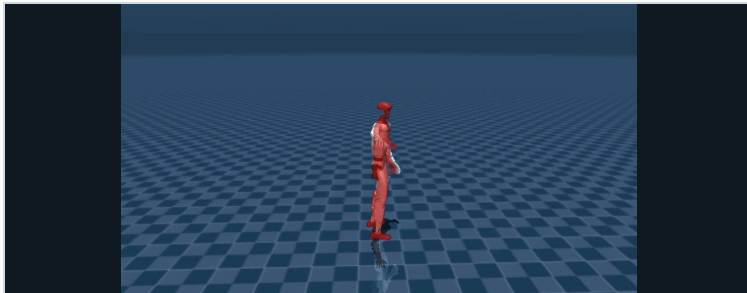
crab_walk

physical pass + SONIC no-fall



wipe_table

physical pass + SONIC no-fall



point_right

physical pass + SONIC no-fall



reach_overhead

physical pass + SONIC no-fall

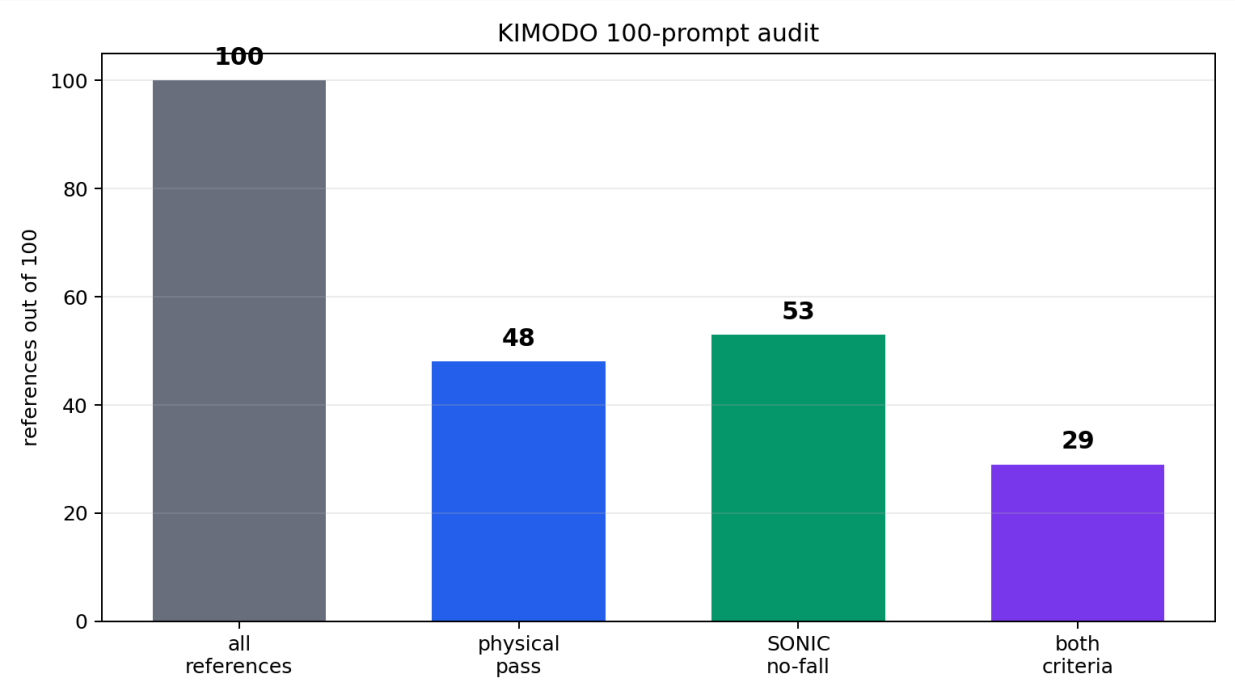


single_leg_balance

physical pass + SONIC no-fall

Six generated references pass the physical rules and complete SONIC immediately. Red ghost = reference; solid G1 = controller rollout.

First-Pass KIMODO Audit

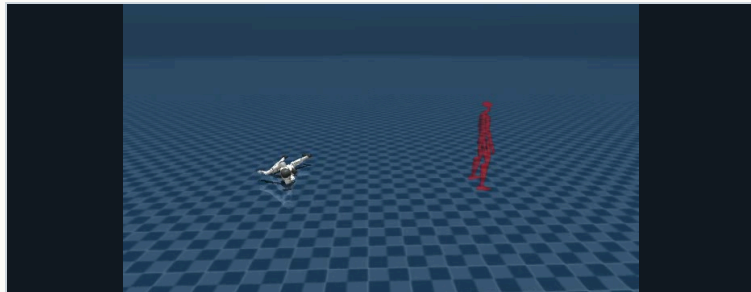


Set	Physical Pass	SONIC No-Fall	Mean SONIC Sec.	Mean RMSE
all KIMODO refs	48/100	53/100	2.855	0.156
physical-pass subset	48/48	29/48	3.293	0.170
flagged subset	0/52	24/52	2.450	0.143

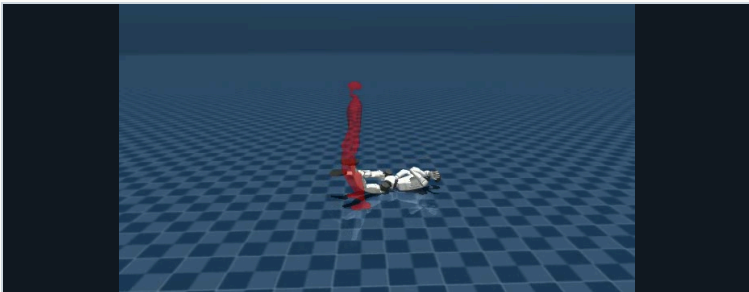
Controller check: 29/48 physical-pass references also complete SONIC; flagged references complete less often and track for less time.

Full 100-prompt KIMODO run. Failure tags identify references that need repair or rejection before deployment.

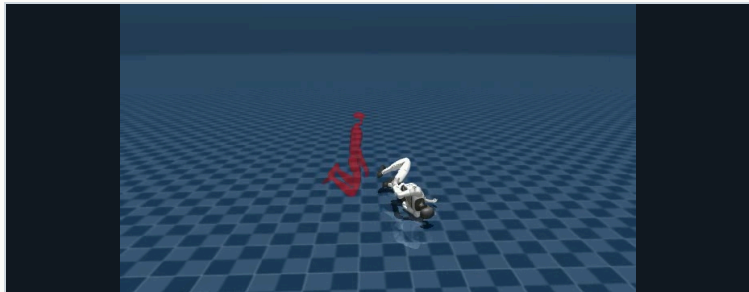
Generated Failure Modes



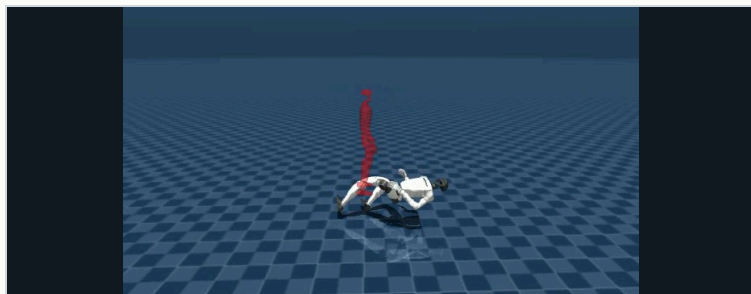
Torque/root wrench
forward_roll, 16.7x



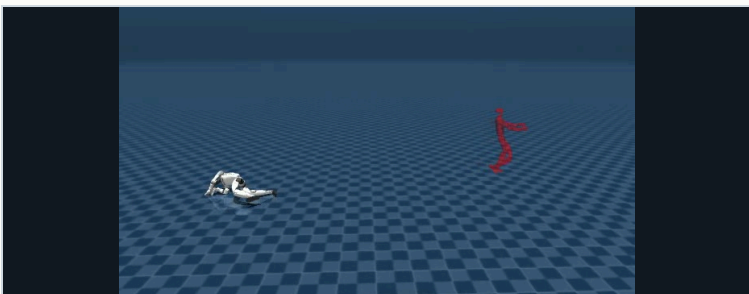
Self-contact
turn_in_place



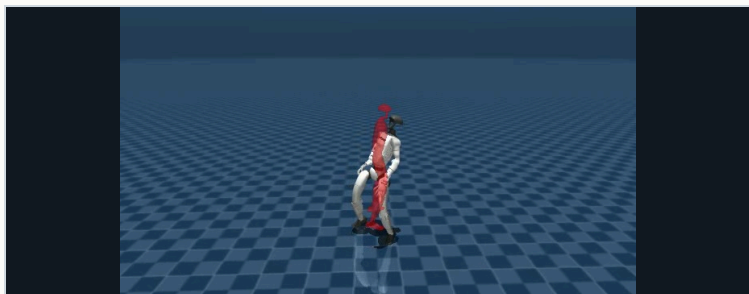
Non-foot floor
knee_slide



Support proxy
step_over_right



Trackability
carry_box



Contact artifact
happy_dance

KIMODO references replayed through SONIC. Red ghost = reference; solid G1 = rollout. Failure labels come from metrics.

Failure Stats Across 100 KIMODO Clips

52 physical-rule fail	47 SONIC fail	66 torque limit >1x
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Failure flags are non-exclusive: one clip can violate actuation, contact, and controller tracking.

physical-rule fail	52
SONIC fail	47
torque limit >1x	66
high root force >5kN	28
self-contact >8%	34
contact artifact >0.45	15
non-foot floor contact	11
floor penetration >8cm	10
low foot support	7

Deterministic Repair Helps Some Clips

SONIC no-fall

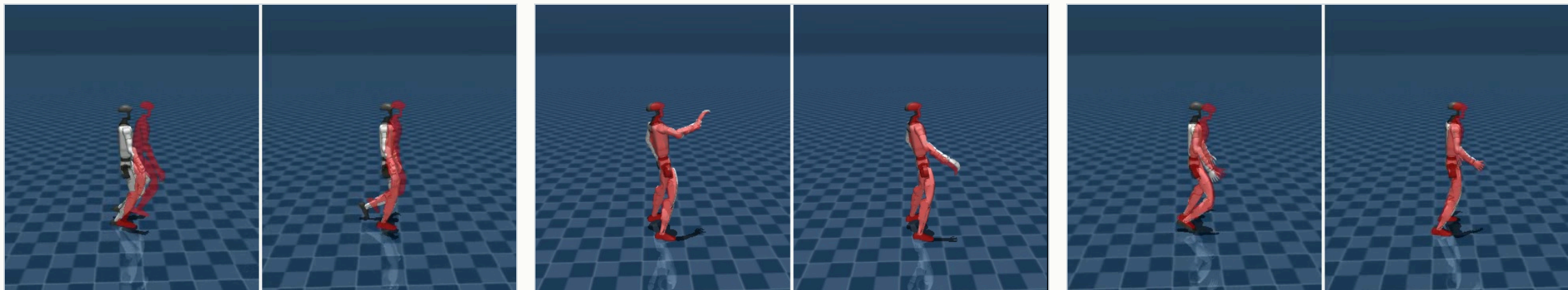
53 → 56

7 rescues, 4 regressions in the repair pass

physical pass	48 → 53	+5	100 clips
critic accept	47 → 54	+7	100 clips
reject / regenerate	18 → 16	-2	100 clips
mean risk	41.5 → 37.9	-3.6	lower is better
mean RMSE	0.156 → 0.142	-0.014	lower is better

Repair baseline: deterministic retiming/smoothing variants selected by the same physical-awareness critic and SONIC gate.

Real Repair Rescues



forward_walk

caught: trackability

prompt: slower, smoother steps

wave

caught: torque / tracking

prompt: planted feet, smooth arm

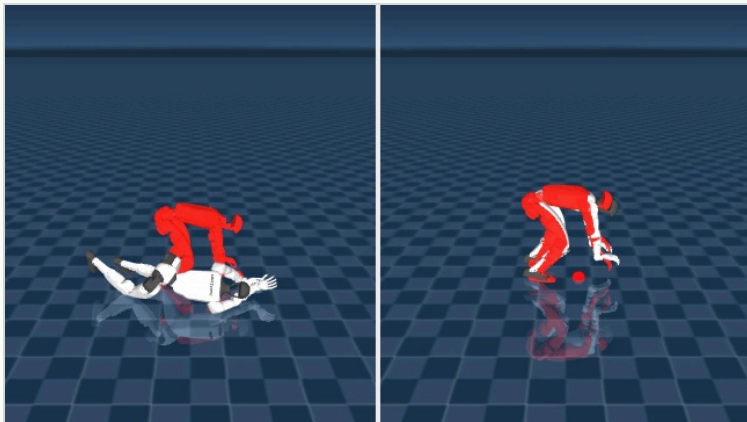
split_squat_jump

caught: support / torque

prompt: smaller jump, soft landing

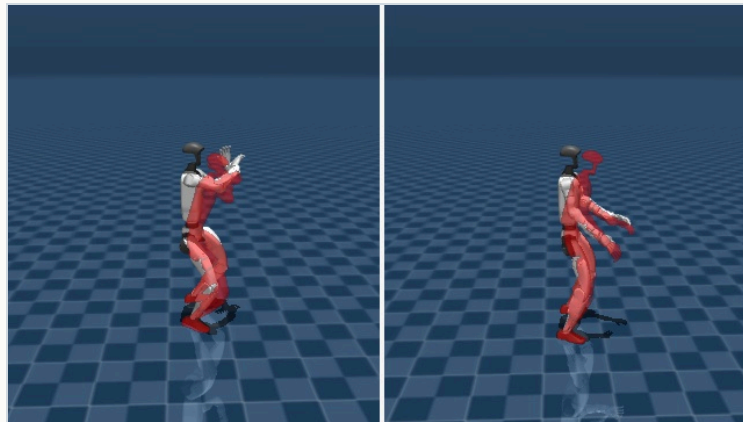
Three representative repair previews. Left side is original KIMODO, right side is repaired. Within each panel: red ghost = reference; solid G1 = physics rollout.

Additional Repair Evidence



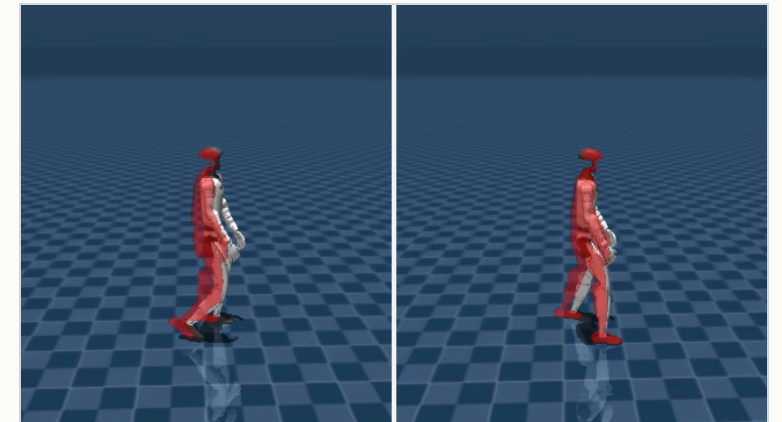
pushup_pose

height-clearance demo
repair pressure: lift torso



happy_dance

risk reduction, still flagged
repair pressure: clean foot placement

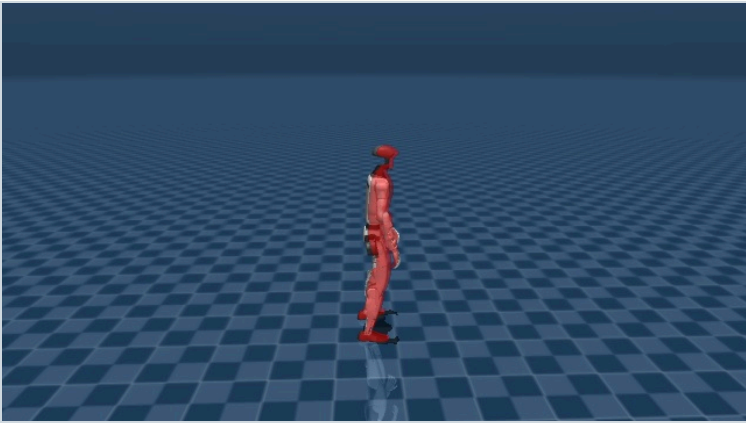


backward_recovery

support-proxy rescue
repair pressure: smaller lean

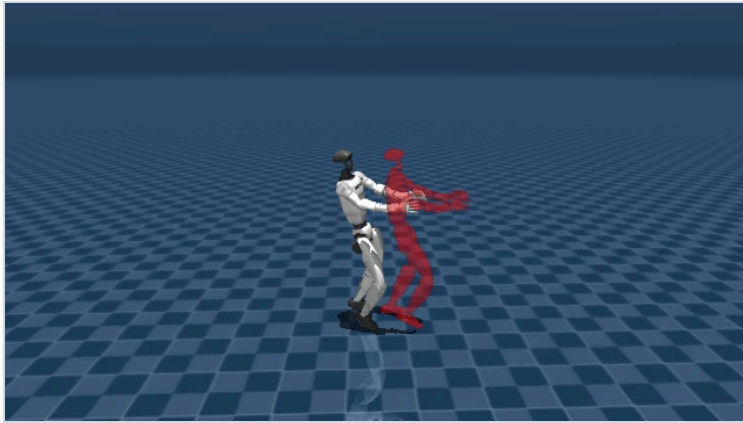
Supporting examples, not all SONIC rescues. Red ghost = reference; solid G1 = rollout or visual repair proxy.

Boundary: What Still Fails



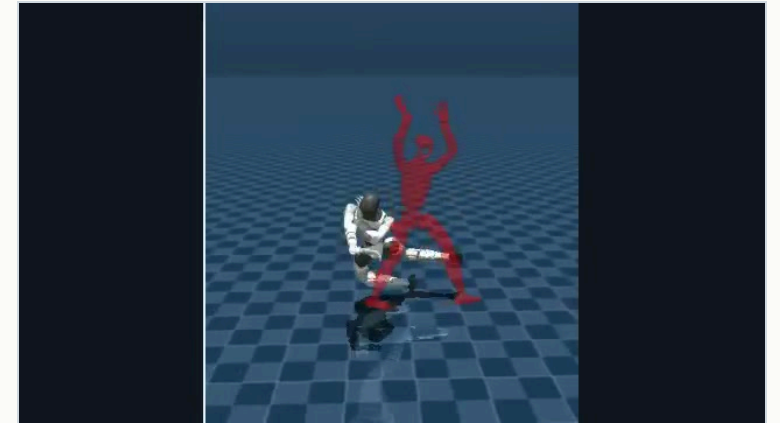
forward_roll

torque/root wrench



carry_box

controller trackability



cartwheel_attempt

torque/root wrench

Still unsolved: arbitrary text-to-robot motion; robust acrobatics and low postures; automatic repair for every invalid reference.

Limitations

Reference repair is useful, not enough

Retiming and smoothing can rescue some clips. Stronger text-to-motion stability likely needs fine-tuning or a learned motion-level refiner.

Tracker-as-foundation is an assumption

We treat SONIC as a foundation motion tracker, but uncovered motions remain a weak point. Dataset coverage still matters.

Trackability is still hard to certify

If a generated reference fails, deciding whether RL could learn to track it is a harder problem than our test-time metric screen.

Iteration has deployment cost

Repair and audit take extra test-time iterations, but catching infeasible motion before controller execution is still important for real-world deployment.

Main point: metric-guided reference repair can reject or improve generated references, but it does not replace better generators, broader tracker training, or deployment-time safety checks.